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Investigation of the acoustic effects of orthographic consistency on Japanese homophones

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Introduction

Background & Research Objectives:

- The phonetic realization of word duration in homophone pairs, such as *time* and *thyme,* systematically differs based on their lexical characteristics, including part of speech, frequency, and orthographic consistency [1].
- Higher orthographic consistency is associated with shorter homophone duration in English [1].
  - Investigate whether orthographic consistency effects on homophone duration generalize beyond alphabetic languages.
  - Examine two directions of orthographic consistency: spelling-to-sound (feedforward) and sound-to-spelling (feedback).

Poster Proceedings

Method

Corpus data: The Corpus of Spontaneous Japanese [2]

- 174 homophone sets (146 pairs, 22 triples, and 6 quadruples), comprising 382 word types and 3,149 word tokens.
  - Median 2 tokens/type; 142 types occur once. Skewed within sets: 最近 (127) / 細菌 (1); 高校 (107) / 後項 (1) / 口腔 (1) / 孝行 (4); 検討 (97) / 見当 (3).
  - 4-mora, 2-logograph words, nouns, and spontaneous monologue.
- Phonological-Orthographic (P-O: sound-to-spelling) and Orthographic-Phonological (O-P: spelling-to-sound) consistencies.

Table 1: Example Japanese P-O consistency index calculation.

Target word	現在	/ge.N.za.i/	'now'	24	
Phonological neighbours	**現代**	/ge.N.**da**.i/	'modern times'	38	OF (Both PN and ON)
	存在	/**so**.N.za.i/	'existence'	28	OF (Both PN and ON)
	犯罪	/**ha**.N.za.i/	'crime'	35	OE (PN but not ON)
	限界	/ge.N.**ka**.i/	'limit'	31	OE (PN but not ON)
Consistency Index	$\dfrac{24+38+28}{24+38+28+35+31}=0.58$				

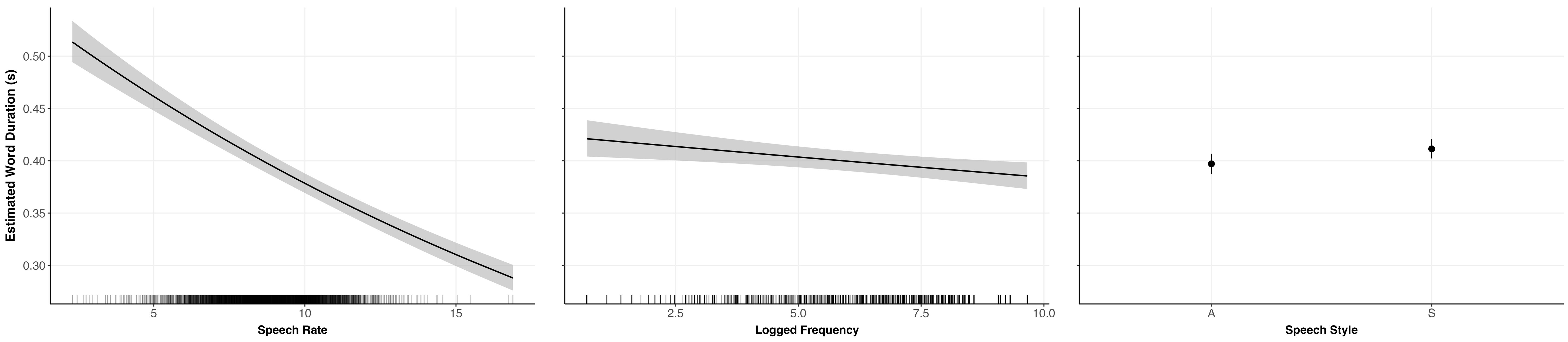
Analysis:

- Dependent variable: Log-transformed Word Duration. Independent variables: *P-O-within*, *O-P-within*.
  - P-O- & O-P-within: each homophone member's consistency value minus its set mean; captures effects among members within each set.
- Covariates: *P-O-between*, *O-P-between*, Speech Rate, Log-transformed Frequency, Log-transformed Bigram Probability, Speech Style.
  - P-O- & O-P-between: mean consistency value of each set; captures effects between sets.

Results & Discussion

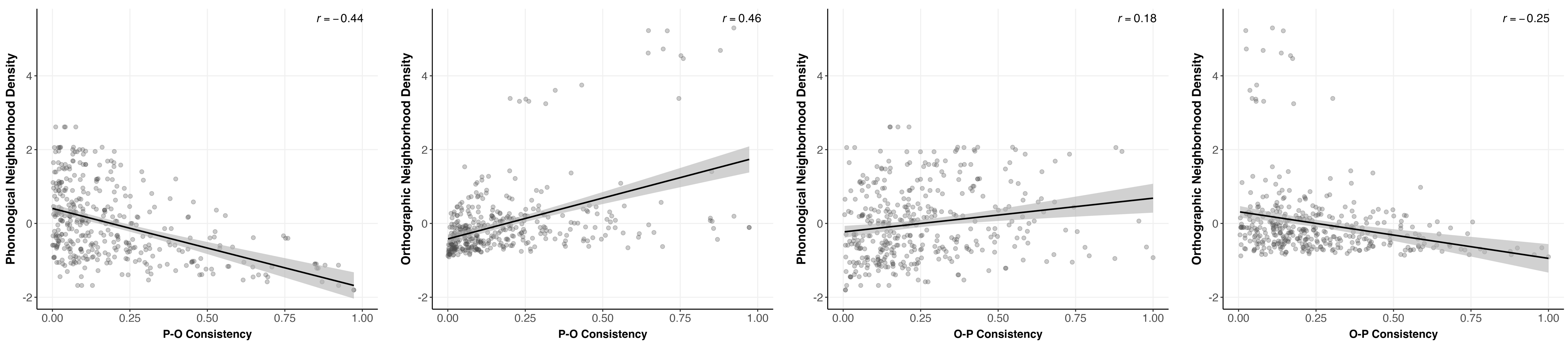
- No effect of orthographic consistency in any direction (O-P: spelling-to-sound or P-O: sound-to-spelling) or level (within or between homophone sets).
- Large effect of Speech Rate; modest effects of Log-transformed Frequency and Speech Style (**Fig. 1**).

Fig. 1: Partial effects of control variables on word duration estimated by the final model.



- The absence of the effect may be due to the distribution and range of consistency in our data set:
  - Limited range of orthographic neighbourhood sizes and clustered consistency values (**Fig. 2**). The contrast is narrower for within-set comparisons.

Fig. 2: Correlations between P-O and O-P consistency and neighbourhood size (n = 382 word types). r values are Pearson correlations.



- Restricted homophone sets with within-set consistency difference above 0.4 (26 sets, 686 tokens):
  - Higher O-P consistency is associated with longer duration (**Fig. 3**).
  - Consistent with Ziegler et al. [3]: dense PNs inhibit production while dense ONs facilitate it.
  - Higher O-P consistency is modestly associated with dense PNs (strong inhibition) and sparse ONs (weak facilitation), resulting in overall inhibition.
- Gahl [1] shows the opposite: higher orthographic consistency (O-P consistency), shorter duration.
- Reversal may stem from how consistency is calculated: grapheme probabilities vs. number and frequency of PNs and ONs.

The boundary between spoken and written language is blurring: text is increasingly produced by speaking and consumed by listening, which may strengthen or weaken orthography's effect on speech.

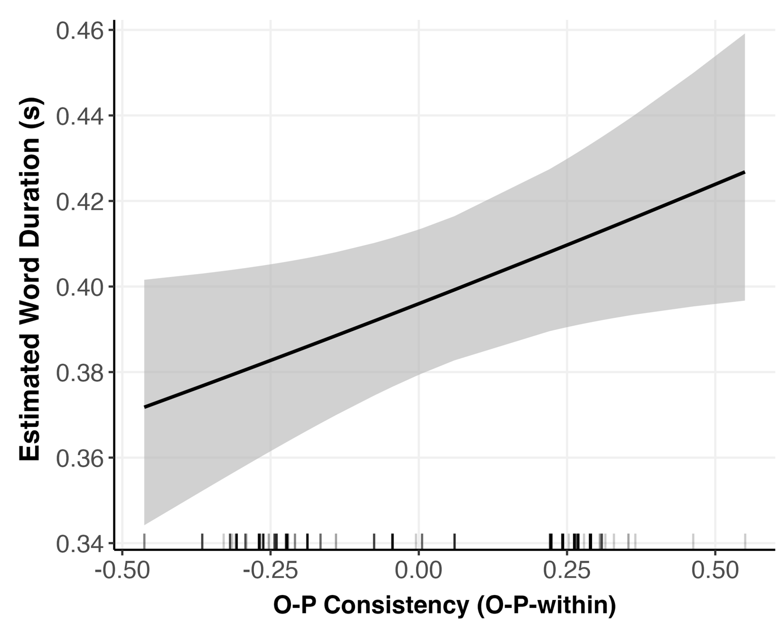


Fig. 3: Partial effects of O-P Consistency (O-P-within) on word duration estimated by the model.

References:
[1] Gahl, S. (2008). Time and thyme are not homophones: The effect of lemma frequency on word durations in spontaneous speech. *Language, 84*(3), 474–496.
[2] Maekawa, K. (2003). Corpus of Spontaneous Japanese: Its design and evaluation [Paper presentation]. ISCA & IEEE Workshop on Spontaneous Speech Processing and Recognition, Tokyo, Japan.
[3] Ziegler, J. C., Muneaux, M., & Grainger, J. (2003). Neighborhood effects in auditory word recognition: Phonological competition and orthographic facilitation. *Journal of Memory and Language, 48*(4), 779–793.

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